

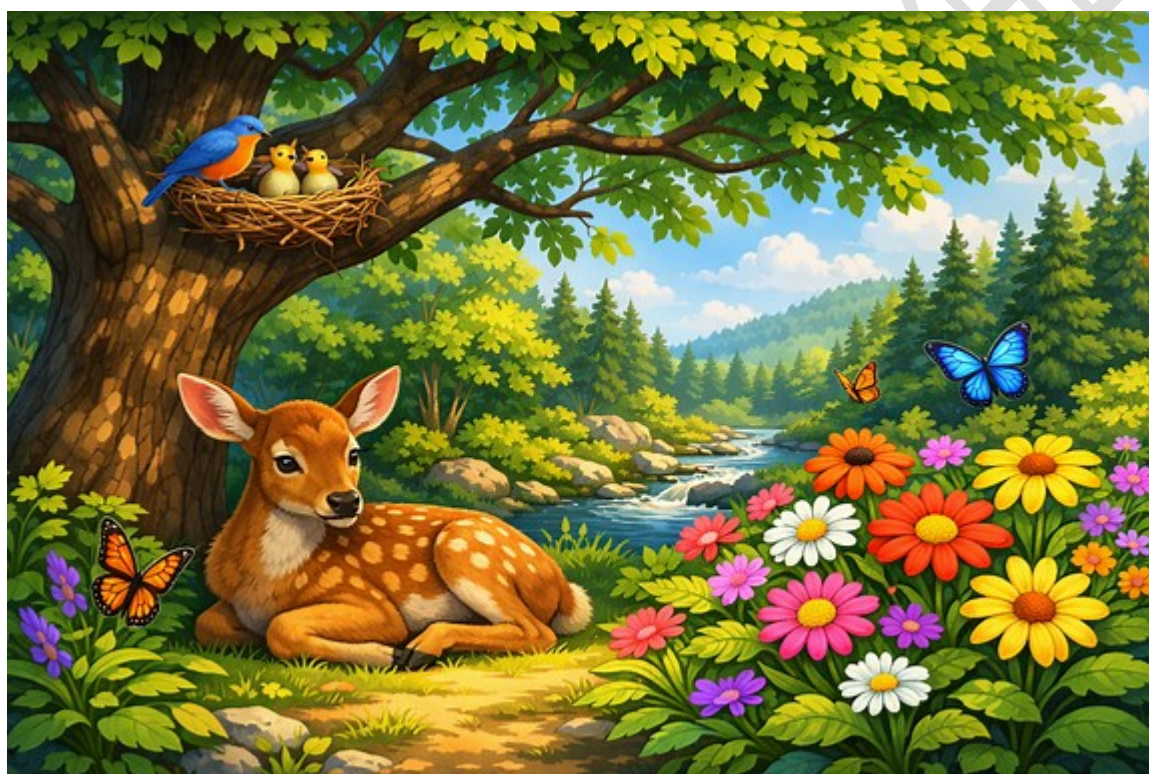
CHAPTER 2:- DIVERSITY IN THE LIVING WORLD

Topic 1: Diversity in Plants and Animals Around Us

Definition Biodiversity refers to the immense variety of plants and animals found in a particular region.

Concept Explanation When you take a nature walk, you will notice that the world is filled with living creatures that look and behave differently. Plants range from tiny grasses to massive trees, and animals range from small ants to large cows. Every single plant and animal plays a unique role in nature; for instance, trees provide food and shelter to birds, while animals help spread seeds. This shows that plants and animals are deeply dependent on each other to survive.

Diagram



Real-Life Example Just by listening to a forest, you can experience biodiversity. Each bird species has its own completely unique chirp and call.

☒ Quick Revision Checklist:

- [] Biodiversity is the variety of life in a specific area.
- [] Every living member plays a specific role in nature.
- [] Plants and animals are interdependent.

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Topic 2: How to Group Plants?

Definition Grouping is the method of arranging plants into categories (like herbs, shrubs, and trees) based on similarities and differences in their physical features.

Concept Explanation Because there are so many plants, we categorize them based on their height, stem thickness, and where their branches start.

- **Herbs:** Typically small plants with soft, green, and tender stems.
- **Shrubs:** Medium-sized plants with hard, woody stems that start branching very close to the ground. Their stems are hard but not as thick as a tree.
- **Trees:** Tall plants with hard, thick, brown woody stems. Their branches start higher up on the stem, away from the ground.
- **Climbers & Creepers:** Plants with very weak stems. Climbers need support to grow upward, while creepers spread out along the ground.

Diagram



Real-Life Example In a garden, a tomato plant is an herb, a rose bush is a shrub, and a mango tree is a tree.

☑ Quick Revision Checklist:

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- [] Grouping makes studying plants easier.
- [] Herbs have soft, green stems.
- [] Shrubs have hard stems branching near the base.
- [] Trees are tall with thick, woody trunks.
- [] Climbers take support; creepers spread on the ground.

Topic 3: Leaf Venation and Root Systems

Definition

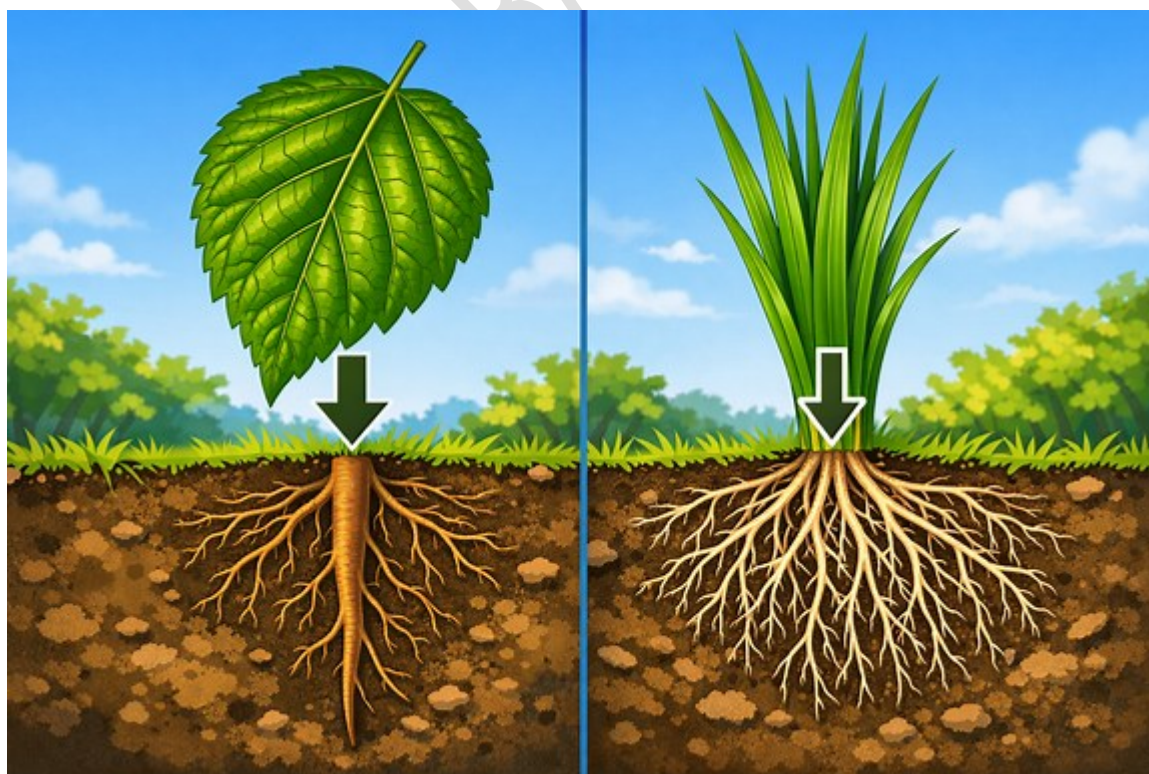
Venation is the pattern of veins on a leaf. A **Taproot** system has one main thick root with smaller side roots, while a **Fibrous root** system is a bunch of similar-sized thin roots arising from the stem's base.

Concept Explanation

Leaves and roots are deeply connected! Leaves show two main vein patterns:

1. **Reticulate Venation:** A net-like pattern on both sides of a thick middle vein.
2. **Parallel Venation:** The veins run straight and parallel to each other. There is a magic rule in nature: Plants with reticulate venation almost always have taproots. Conversely, plants with parallel venation almost always have fibrous roots.

Diagram



Real-Life Example If you pull up a mustard plant or a chickpea plant, you will see a main taproot, and its leaves will have net-like veins. If you look at common grass or wheat, the leaves have parallel lines, and the roots look like a messy mop (fibrous).

☑ Quick Revision Checklist:

- [] Venation is the design of veins on a leaf.
- [] Reticulate = net-like; Parallel = straight lines.
- [] Taproot = one main root; Fibrous = bunch of thin roots.
- [] Rule: Reticulate Venation $\rightarrow$ Taproot.
- [] Rule: Parallel Venation $\rightarrow$ Fibrous Root.

Topic 4: Types of Seeds (Dicots and Monocots)

Definition

Cotyledons are the seed leaves or the split parts inside a seed. **Dicots** are plants with two cotyledons, and **Monocots** are plants with a single cotyledon.

Concept Explanation If you soak a seed in water and remove its coat, you will see its internal structure. Some seeds naturally split into two equal halves; these are called dicotyledons (dicots). Other seeds do not split and remain as a single thin piece; these are monocotyledons (monocots). This links back to our previous rule to form a complete picture:

- **Dicot plants** = 2 Cotyledons + Reticulate Venation + Taproot System.
- **Monocot plants** = 1 Cotyledon + Parallel Venation + Fibrous Root System.

Diagram



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Real-Life Example Chickpeas (chana) and kidney beans can easily be split into two halves with your fingers after soaking, making them dicots. A kernel of maize (corn) cannot be split perfectly in half, making it a monocot.

☑ Quick Revision Checklist:

- ☐ Cotyledons are the internal parts of a seed.
- ☐ Dicots = 2 halves. Features: Reticulate veins, Taproots.
- ☐ Monocots = 1 piece. Features: Parallel veins, Fibrous roots.

Topic 5: How to Group Animals?

Definition Animals can be grouped based on their types of movement and the specific body parts they use to travel from one place to another.

Concept Explanation Just like plants, the animal kingdom is vastly diverse. To make studying them easier, we categorize them by how they navigate their environment. Animals can fly, run, crawl, walk, hop, or jump. To achieve this, they use specialized body parts like wings, legs, and fins. We can also group animals based on shape, size, structure, and color.

Diagram



Real-Life Example A pigeon uses its legs to walk on the ground and its wings to fly in the air. A fish uses its fins to swim gracefully through water.

☑ Quick Revision Checklist:

- ☐ Animals are grouped by movement (walk, fly, crawl, swim).

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- [] Different movements require different body parts (legs, wings, fins).

Topic 6: Adaptations and Habitats

Definition

Adaptations are the special features that enable plants and animals to survive in a particular region. A **Habitat** is the specific environment or place where these plants and animals live.

Concept Explanation Biodiversity changes depending on where you are on Earth because the environmental conditions change. Plants and animals develop special bodily features (adaptations) to handle their specific climate. Habitats are generally divided into two main categories:

1. **Terrestrial Habitats:** Plants and animals that live on land (forests, deserts, grasslands, mountains).
2. **Aquatic Habitats:** Plants and animals that live in water (ponds, lakes, rivers, oceans).
Note: Animals like frogs that can survive in both are called amphibians. Protecting these habitats is crucial. When a habitat is damaged by human activity, animals lose their homes and food, leading to a dangerous loss of biodiversity.

Diagram



Real-Life Example

- **Deserts:** Hot desert camels have long legs and wide hooves so they don't sink into the sand, and they store fat in a single hump. Cold desert camels have shorter legs, two humps, and long hair for warmth. Cactus plants have thick, fleshy stems to store water.

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- **Mountains:** Deodar trees are conical with sloping branches so that snow slides right off them.

☒ Quick Revision Checklist:

- ☐ Adaptations are special survival features.
- ☐ Habitats provide food, water, air, and shelter.
- ☐ Terrestrial = Land-based (deserts, mountains).
- ☐ Aquatic = Water-based (oceans, rivers).
- ☐ Amphibians live on both land and water.
- ☐ Protecting habitats prevents the loss of biodiversity.